

Solar cell fabric to recharge phones on-the-go

Japanese researchers have created a solar-cell fabric that can harness solar energy while the wearer is on the go. The fabric is made of wafer-thin solar cells woven together. It opens the door to the possibility of people powering up their mobile phones and other electronics gadgets with their sweaters or trousers.



Solar cell fabric (Courtesy: www.straittimes.com)

"We still have things to solve before commercialisation, such as coating for the conductive wires and improving the fabric's durability," said an official at the Industrial Technology Center in central Japan's Fukui Prefecture. "But we have already been contacted by electronics makers, blind makers and others who showed interest in our invention," he added.

The centre developed the fabric in association with a Kyoto-based solar cell maker and other private firms.

Waterless solar panel cleaning device

Georg Eitelhuber, a physics teacher at the King Abdullah University of Science and Technology (KauST) in Saudi Arabia, has created a waterless solar panel cleaning device for use in harsh desert environments like that of Saudi Arabia. Called Nomadd, the device makes use of an automated 'dry sweep' to push dust and dirt away from surfaces of the solar panel. It can even clean the thick dust that sticks after a rain shower or sand storm in humid conditions—according to a report by BBC.

Nomadd is claimed to be rugged and low-maintenance with just four major moving parts. The device can run along a bank of solar panels, sweeping away the dust. Powered by lithium-ion batteries, it is charged by the array and can manoeuvre over gaps and obstacles between the panels. It can be used automatically on a schedule.

4-D transistor

Researchers from Purdue and Harvard Universities have designed a transistor that appears to be a Christmas tree. This 4-D transistor is a preview of things to come in the semiconductor industry.

The transistor contains three nanowires made from indium-gallium-arsenide—a material that could replace silicon within a decade. The three nanowires are progressively smaller, yielding a tapered cross-section resembling a Christmas tree.

The 4-D transistor may not only replace silicon but also make way for faster and more efficient integrated circuits. With this technology, the weight of your laptop will reduce considerably and it will not heat-up easily either.

The length of gates in current 3D transistors is about 22 nanometres. Researchers are working to make them as small as 10 nanometres.

Explaining on the 4-D aspect of transistors, Peide "Peter" Ye, a professor of electrical and computer engineering at Purdue University, said, "A one-story house can hold so many people, but more floors, more people, and it's the same thing with transistors. Stacking them results in

more current and much faster operation for high-speed computing. This adds a whole new dimension, so I call them 4-D."

A new spam filter

Concordia University scientists have developed a more efficient and effective spam filter than those being used presently.

Ola Amayri, a researcher, told PTI: "Our new method for spam filtering is able to adapt to the dynamic nature of spam e-mails and accurately handle spammers' tricks by carefully identifying informative patterns, which are automatically extracted from both text and images content of spam e-mails."

As of now, the focus was on automatic extraction of spam mails from the mail box but now the plan is to use these tricks in combination. Most spam filters target only the content or an image, based on which they automatically mark a mail as spam. But, Amayri said, the new proposed statistical framework will consider the pattern of not just text but images as well, and that too simultaneously.

Indian robot controlled by smartphone

Robots Alive has developed Andy—a personal smart robot that can be controlled by a mobile phone connected to it. The mobile phone communicates with the robot through the audio cable. Therefore any software which can run on the mobile phone and utilise the audio, can be programmed to drive the robot.

Abheek Bose, Andy's creator, said, "Personal robots till date have been unaffordable due to the high costs of the hardware involved. However, we fail to see that the same hardware exists in the smart devices available today. Utilising the power of these devices, we can make intelligent robots highly affordable for individuals!"

Andy has been designed to educate young students on robotics and electronics. Developers can create their own apps to enable Andy with more features.



A pile of transistors (Courtesy: www.crazyengineers.com)

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